Matrices, Inverse matrices by elementary row

- **Matrices**
- When a system of equations has more than two equation, it is more convenient to use matrices and vectors in solution.
- The size of the matrix is described by the number of its rows and columns. A matrix of n rows and m columns is represented by $(n \times m)$ matrix.
- $A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1m} \\ a_{21} & a_{22} & \cdots & a_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nm} \end{bmatrix}_{n\times m}$ $, i = 1,2, \dots, n;\ \ j = 1,2, \dots, m$
- **Types of matrices:**
- **Square matrix** : it is a matrix that includes number of rows equals to number of columns $(n = m)$
- $A = \begin{bmatrix} 1 & 4 \\ 2 & 5 \end{bmatrix}_{2\times 2}$ $, B = \begin{bmatrix} 1 & 3 & 0 \\ 3 & 2 & 1 \\ 1 & 8 & 0 \end{bmatrix}_{3\times 3}$
- **Diagonal matrix**: it is a square matrix which all of its elements are zeros except the elements on the main diagonal.
- $A = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 7 & 0 \\ 0 & 0 & 2 \end{bmatrix}$
- Identity matrix: it is a diagonal matrix but the elements on the main diagonal are equal to 1 and it is denoted by I_n
- $I_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$
- Transpose matrix: transpose of A is denoted by A^T which means writing the rows of A as columns in A^T.
- $A = \begin{bmatrix} 9 & 6 & 5 \\ 1 & 2 & 4 \end{bmatrix}_{2\times 3}$ $, A^T = \begin{bmatrix} 9 & 1 \\ 7 & 2 \\ 5 & 4 \end{bmatrix}_{3\times 2}$

- Matrix addition : if $A = [a_{ij}], B = [b_{ij}]$ and both $A\&B$ are $m \times n$ matrices, then
- $A + B = [a_{ij}] + [b_{ij}] = [a_{ij} + b_{ij}]$
- EX : $\begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix} + \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix} = \begin{bmatrix} 3 & 2 \\ 4 & 7 \end{bmatrix}$

Note : for any scalar (number) c, it can multiply the matrix A by c as follows:

$$cA = c[a_{ij}] = [ca_{ij}]$$

Ex: $3\begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 3 & -3 \\ 0 & 6 \end{bmatrix}$

- **Note:**
- The matrix with only one column, $m \times 1$ in size is called a column vector, while with only one row, $1 \times n$ in size is called a row vector.
- Matrix multiplication: let A be $m \times k$ matrix and B be $k \times n$ matrix then $C = AB$ is an $m \times n$ matrix, where
- $C_{ij} = \sum_{t=1}^{k} a_{it} b_{tj} \quad ; i = 1,2, \dots, m \, , j = 1,2, \dots, n$
- Ex: $A = \begin{bmatrix} 3 & 7 & 1 \\ -2 & 1 & -3 \end{bmatrix}_{2\times3}, B = \begin{bmatrix} 5 & -2 \\ 0 & 3 \\ 1 & -1 \end{bmatrix}_{3\times2}$
- $AB = \begin{bmatrix} 16 & 14 \\ -13 & 10 \end{bmatrix}_{2\times2}$
- $BA = \begin{bmatrix} 19 & 33 & 11 \\ -6 & 3 & -9 \\ 5 & 6 & 4 \end{bmatrix}_{3\times3}$

Gauss Jordan elimination method

- In the Gaussian elimination method, we write simpler equivalent augmented matrices, where each row of an augmented matrix represents an equation that can perform the row operations on the augmented matrix.
- Steps:
  1) Construct the augmented matrix $(A: B)$.
  2) Applying row operations including (adding or subtracting two rows, interchange two rows, multiplying any row by any constant except zero.
- Let A be a matrix, X a column vector, B a column vector then the system of linear equations is denoted by $AX = B$
- The solution to a system of linear equations starts by the augmented matrix as shown for the following system:
- $x - 2y = -5$
- $3x + y = 6$
- **Note: number of variables equals to the number of equations**
- Depends on the coefficients of x, y and the constants on the right-hand side of the equation. The matrix of coefficients for this system 2×2 matrix. $\begin{bmatrix} 1 & -2 \\ 3 & 1 \end{bmatrix}$
- If we insert the constants from the right-hand side of the system into matrix of coefficients, we get the 2×3 matrix. $\left[\begin{array}{cc|c} 1 & -2 & -5 \\ 3 & 1 & 6 \end{array}\right]$
- We use a vertical line between the coefficients and constants to represent the equal signs. This matrix is augmented matrix of the system also it can be written as: $\begin{bmatrix} 1 & -2 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -5 \\ 6 \end{bmatrix}$
- **Application of Gauss Jordan elimination method to solve the system $(\boldsymbol{AX = B})$**

EX1 : use Gaussian elimination method to solve the following system of equations

- $x - 3y = 11$
- $2x + y = 1$
- Sol: (Note : no. of variables = no. of equations = 2)
- The augmented matrix :
- $\left[\begin{array}{cc|c} 1 & -3 & 11 \\ 2 & 1 & 1 \end{array}\right]$

Now we want to get this matrix $\left[\begin{array}{cc|c} 1 & 0 & x \\ 0 & 1 & y \end{array}\right]$ by applying row operations as follows:

- $\left[\begin{array}{cc|c} 1 & -3 & 11 \\ 2 & 1 & 1 \end{array}\right] \rightarrow R_2' = -2R_1 + R_2 \rightarrow \left[\begin{array}{cc|c} 1 & -3 & 11 \\ 0 & 7 & -21 \end{array}\right] \rightarrow R_2' = \frac{1}{7}R_2$
- $\left[\begin{array}{cc|c} 1 & -3 & 11 \\ 0 & 1 & -3 \end{array}\right] \rightarrow R_1' = 3R_2 + R_1 \rightarrow \left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & -3 \end{array}\right]$
- The solution to the system is $(X = 2\ and\ y = -3)$
- To check the result, substitute the values of $x\&y$ in any equation, such as in $(x - 3y = 11)$

$2 - 3(-3) = 11$

$2 + 9 = 11$

$11 = 11 \rightarrow L.H.S = R.H.S$

Ex2: use Gaussian elimination method to solve the following system of equations

- $2x - y + z = -3$
- $x + y = -z = 6$
- $3x - y - z = 4$
- $\left[\begin{array}{cccc} 2 & -1 & 1 & -3 \\ 1 & 1 & -1 & 6 \\ 3 & -1 & -1 & 4 \end{array}\right]$ → we want to get this matrix $\left[\begin{array}{cccc} 1 & 0 & 0 & x \\ 0 & 1 & 0 & y \\ 0 & 0 & 1 & z \end{array}\right]$
- $R_1 \leftrightarrow R_2 \rightarrow \left[\begin{array}{cccc} 1 & 1 & -1 & 6 \\ 2 & -1 & 1 & -3 \\ 3 & -1 & -1 & 4 \end{array}\right]$,
- $R_3' = -3R_1 + R_3 \& R_2' = -2R_1 + R_2 \rightarrow \left[\begin{array}{cccc} 1 & 1 & -1 & 6 \\ 0 & -3 & 3 & -15 \\ 0 & -4 & 2 & -14 \end{array}\right]$
- $R_2' = -\frac{1}{3}R_2 \rightarrow \left[\begin{array}{cccc} 1 & 1 & -1 & 6 \\ 0 & 1 & -1 & 5 \\ 0 & -4 & 2 & -14 \end{array}\right]$
- $R_1' = -R_2 + R_1 \& R_3' = 4R_2 + R_3 \rightarrow \left[\begin{array}{cccc} 1 & 1 & 0 & 1 \\ 0 & 1 & -1 & 5 \\ 0 & 0 & -2 & 6 \end{array}\right]$
- $R_3' = \frac{1}{2}R_3 \rightarrow \left[\begin{array}{cccc} 1 & 1 & -1 & 6 \\ 0 & 1 & -1 & 5 \\ 0 & -4 & 2 & -14 \end{array}\right]$
- $R_2' = R_3 + R_2 \rightarrow \left[\begin{array}{cccc} 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 2 & -3 \end{array}\right]$
- The solution to the system is $(x = 1, y = 2 \; and \; z = -3)$
- To check the result, substitute the values of $x = 1, y = 2 \; \& z = -3$ in any equation, such as in equation $(2x - y + z = 3)$

$2(1) - 1(2) + (-3) = -3$

$2 - 2 - 3 = -3 \qquad \therefore -3 = -3 \rightarrow L.H.S = R.H.S$

Homework: solve the following equations by Gauss-Jordan Elimination Method and check the result.

1) $2x - y = 18$

$3x + y = 2$

2) $3x - 2y + 8z = 9$

$-2x + 2y + z = 3$

$x + 2y - 3z = 8$

Matrices, Inverse matrices by elementary row

- **Determinant**
- Determinant is a value that can be calculated from the elements of a **square matrix.** The determinant of a matrix A is denoted $\det(A)$, or the determinant of A.
- It used to find the inverse of a matrix and useful in calculus for several application.
- It used to check whether or not a matrix can be inverted, there if $det(A) = 0$

Then there is no inverse.

- The calculation of determinant is as follows:

For 2×2 matrix,

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}, \det\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21}$$

Ex1: find the determinant of *A*

$$A = \begin{bmatrix} 1 & 3 \\ -2 & 5 \end{bmatrix} = \begin{vmatrix} 1 & 3 \\ -2 & 5 \end{vmatrix} = 1(5) - 3(-2) = 5 + 6 = 11$$

For 3×3 matrix,

To each element of a 3×3 matrix there corresponds a 2×2 matrix that is obtained by deleting the row and column of that element. The determinate of the 2×2 matrix is called the minor of that element.

- $$det\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

$$= a_{11}\begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12}\begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{12}\left|\begin{bmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix}\right|$$

Ex2: find the determinant of *A*

$$\begin{vmatrix} 3 & 8 & 1 \\ 6 & 2 & -1 \\ -1 & -4 & 1 \end{vmatrix} = 3\begin{vmatrix} 2 & -1 \\ -4 & 1 \end{vmatrix} - 8\begin{vmatrix} 6 & -1 \\ -1 & 1 \end{vmatrix} + \begin{vmatrix} 6 & 2 \\ -1 & -4 \end{vmatrix}$$

$$= 3(2-4) - 8(6-1) + (-24+2) = -68$$

- Notice the $+ - +$ pattern for the numbers of the first row.

- **Finding the inverse of matrices by elementary row method**: Also called Gauss Jordan elimination method.
- Construct the augmented matrix $(A: I)$
- Using row operations: Change the rows using :1) Adding or subtracting the row by another row. 2) Multiplying the row by a constant and 3)Swapping rows until convert matrix A into the Identity Matrix I, $(I: A^{-1})$

Note:

1) Augmented matrices appear in Linear algebra as two appended matrices and are useful for solving systems of linear equations.
2) It can check the result through multiplying the original matrix by the inverse matrix to get the identity matrix $(AA^{-1} = I)$.

EX1: Find A^{-1} using elementary row method (Gaussian elimination)

- $A = \begin{bmatrix} 2 & 1 \\ 1 & 4 \end{bmatrix} \rightarrow \left[\begin{array}{cc|cc} 2 & 1 & 1 & 0 \\ 1 & 4 & 0 & 1 \end{array}\right] \rightarrow R_1 = \frac{1}{2}R_1 \rightarrow \left[\begin{array}{cc|cc} 1 & \frac{1}{2} & \frac{1}{2} & 0 \\ 1 & 4 & 0 & 1 \end{array}\right]$
- $R_2' = R_2 - R_1 \rightarrow \left[\begin{array}{cc|cc} 1 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & \frac{7}{2} & \frac{-1}{2} & 1 \end{array}\right], R_2' = \frac{2}{7}R_2 \rightarrow \left[\begin{array}{cc|cc} 1 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 1 & \frac{-1}{7} & \frac{2}{7} \end{array}\right]$
- $R_1' = R_1 - \frac{1}{2}R_2 \rightarrow \left[\begin{array}{cc|cc} 1 & 0 & \frac{4}{7} & \frac{-1}{7} \\ 0 & 1 & \frac{-1}{7} & \frac{2}{7} \end{array}\right], \rightarrow A^{-1} = \begin{bmatrix} \frac{4}{7} & \frac{-1}{7} \\ \frac{-1}{7} & \frac{2}{7} \end{bmatrix}$
- $AA^{-1} = I \rightarrow \begin{bmatrix} 2 & 1 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} \frac{4}{7} & \frac{-1}{7} \\ \frac{-1}{7} & \frac{2}{7} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$